

Assessing ABDM's Design-Reality Gap for Kerala's Interstate Informal Migrant Workforce

A Systems Analysis of Continuity-of-Care Barriers

Research proposal · Kerala State Planning Board Student Internship Programme 2026–27

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Abstract

India's Ayushman Bharat Digital Mission (ABDM) was designed to resolve fragmented health records through a federated architecture built on ABHA digital health IDs, HL7 FHIR interoperability, and consent-driven health information exchange. Most published evaluations of ABDM implicitly assume a stable, single-location patient. Kerala's interstate informal migrant workforce – an estimated 3.5 million individuals in construction, plantation, fisheries, and allied informal-sector work – rarely meets that assumption. This proposal examines whether ABDM's current architecture supports continuity of care for this population, identifies where the design and the lived reality diverge, and proposes a hybrid, privacy-preserving fallback architecture for the cases where it does not. The study is framed as a design-oriented socio-technical systems analysis, combining policy documents, administrative implementation data, and peer-reviewed literature, supported by an open-source architectural prototype.

1. Problem Statement

Interstate informal migrant workers in Kerala experience healthcare in fragments. A worker's medical history, generated in a home state such as West Bengal, Bihar, Odisha, or Assam, does not reliably travel with them to a destination-state clinic or hospital. ABDM was built to solve exactly this kind of fragmentation through ABHA-linked, FHIR-based health information exchange. What remains unclear is whether this architecture, designed around assumptions of stable registration and consistent digital access, actually functions for a population defined by mobility, variable documentation, and uneven digital literacy. This is not primarily a question of whether migrant workers can be persuaded to adopt a digital ID – it is a question of whether the system's design already accounts for them.

2. Rationale and Literature Gap

Existing research occupies three separate corners of this problem, but none occupies their intersection. Gandhi and Soundappan (2024) study ABHA adoption and EHR perception among a stable, urban north Indian population – a useful adoption baseline, but not a mobile, informal-sector population. Ummer et al. (2021) document Kerala's institutional capacity to deploy coordinated digital health infrastructure for migrant and mobile populations during the COVID-19 response – strong evidence of feasibility, but pre-dating ABDM. Bangla, Kapoor, and Jeer (2026) provide a global theoretical framework showing that digital health systems which do not deliberately design for excluded populations tend to reproduce rather than resolve inequity – a strong theoretical lens, but not specific to ABDM or Kerala. This proposal sits in the empty space these three papers leave open.

Layer	Source	Contribution
Policy vision	National Digital Health Blueprint (2019)	Defines ABDM's intended architecture and governance principles
Implementation architecture	Mantri et al. (2024)	Explains ABDM as an operationalization of the WHO Digital Health Platform model
Historical context	Raj, Dananjayan & Agarwal (2023)	Traces the policy evolution from NDHB to NDHM to ABDM
Implementation status	NHA ABDM Public Dashboard	Real-time administrative indicators of adoption
Global systems context	Jayathissa & Hewapathirana (2023)	Interoperability failures across LMICs are structural, not merely technical
Equity theory	Bangla, Kapoor & Jeer (2026)	Frames digital-by-default marginalization; grounds the equity argument
Kerala policy lens	Sreekumar (2023)	Shows that declared policy equity does not guarantee operational equity
Empirical adoption evidence	Gandhi & Soundappan (2024)	ABHA adoption dynamics in a stable, general urban population
Kerala readiness	Ummer et al. (2021)	Demonstrates Kerala's institutional capacity for coordinated digital deployment
Prototype architecture	Ngo et al., RECONNECT (2024)	Federated, privacy-preserving linkage design without a centralized database

Table 1. Literature architecture – how each source is used in this proposal.

3. Research Objectives

- Map ABDM's architecture against the WHO Digital Health Platform reference model, and identify where its design assumptions diverge from the lived conditions of Kerala's interstate informal migrant workforce.
- Apply an integrated digital health equity framework to assess whether interstate informal migrant workers are structurally accounted for within ABDM, or only implicitly folded into the general patient population.
- Design and demonstrate, through an open-source architectural prototype, a hybrid continuity-of-care pathway that remains functional when ABHA-based identification is unavailable or unreliable.

4. Conceptual Framework

This study treats ABDM not as a single application but as a layered digital public infrastructure, following the WHO-ITU Digital Health Platform Handbook. Each layer, from physical infrastructure up to patient-facing applications, depends on the layer beneath it; a gap introduced at a lower layer (for instance, in how the ABHA registry handles unstable addresses) propagates upward into every application built on top of it. Figure 1 maps this reference architecture onto ABDM's actual building blocks.

Figure 1. WHO Digital Health Platform layers mapped to ABDM building blocks



Figure 1. WHO Digital Health Platform layers mapped to ABDM building blocks. Adapted from Mantri et al. (2024) and Raj, Dananjayan & Agarwal (2023).

5. The Design-Reality Gap: Patient Journey Analysis

To make the gap concrete, the study traces a synthetic interstate migrant worker through nine stages of a care journey, identifying at each stage what happens to their health information under current ABDM assumptions.

Stage	What typically happens to the data
Home state baseline	A health record may exist, but is not guaranteed to be ABHA-linked
Migration	No automatic mechanism transfers records across state lines
Destination registration	Without ABHA, a new, disconnected local record is created
ABHA usage	If unavailable, continuity depends entirely on manual history-taking
Clinical consultation / lab / pharmacy	Data is recorded locally but may not link to prior history
Referral	Cross-facility transfer of referral summaries is not guaranteed
Discharge and return	No guaranteed mechanism syncs the destination record back home

Table 2. Nine-stage patient journey and associated continuity risk, synthesized from ABDM architecture documentation and the reviewed literature.

6. Proposed Hybrid Continuity Architecture

Rather than proposing a replacement for ABHA, this study designs a fallback-aware architecture inspired by the RECONNECT project's approach (Ngo et al., 2024) to linking records across Ireland's independent health systems without a centralized database or exposed identifiers. When ABHA is available, the standard ABDM-compliant FHIR-based exchange applies. When it is not, a privacy-preserving record linkage step – matching on encrypted demographic attributes such as name, date of birth, and phone number – provides a fallback path into the same consent-governed federated query layer. Data remains at its source system; only queries travel between systems.

Figure 2. Hybrid continuity-of-care architecture

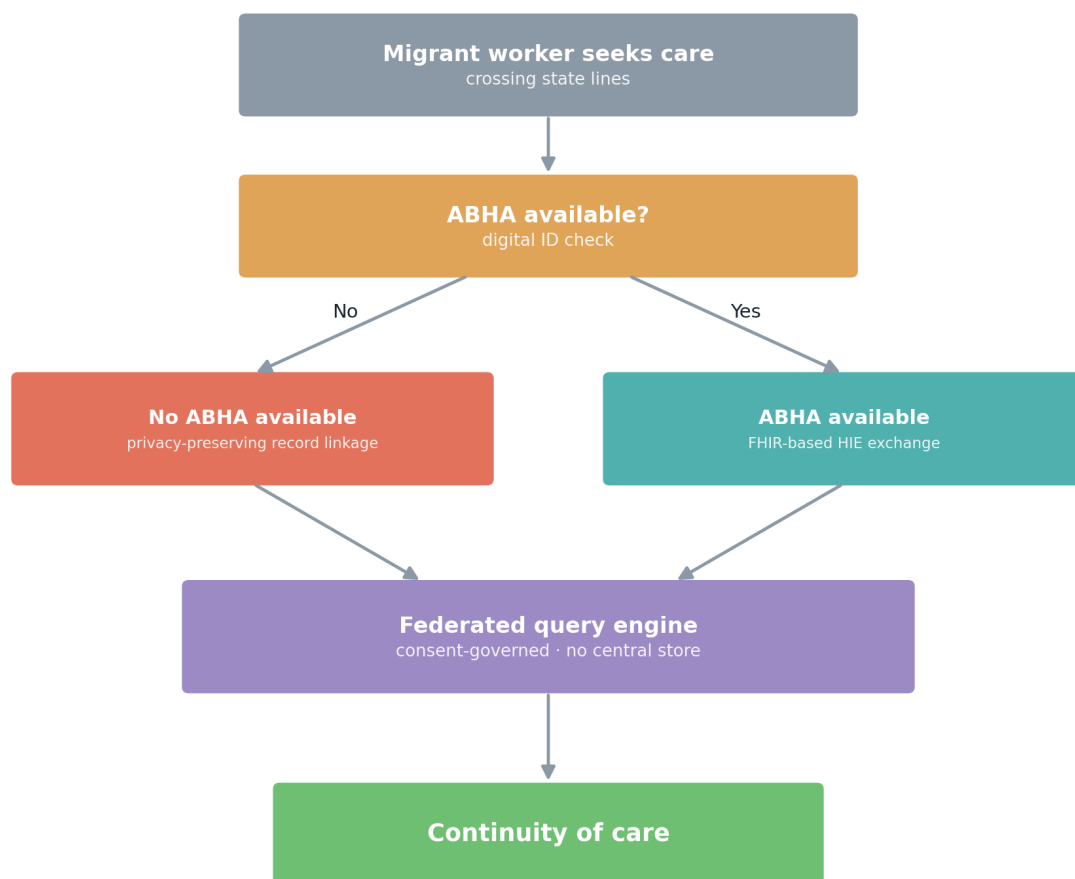


Figure 2. Hybrid continuity-of-care architecture: primary ABHA-based pathway with a privacy-preserving fallback.

7. Methodology

This is framed as a design-oriented socio-technical systems study rather than a systematic review or a primary adoption survey. It combines four categories of source material, each serving a distinct methodological purpose.

Source category	Examples	Purpose
Primary policy documents	National Digital Health Blueprint (2019), ABDM Strategy Overview, Health Data Management Policy	Policy analysis – what the system is intended to do
Primary administrative data	NHA ABDM Public Dashboard	Implementation indicators – what has actually been deployed
Primary technical standards	HL7 FHIR, ABDM FHIR Implementation Guide (NRCES)	Technical framework for the prototype
Peer-reviewed literature	Sources listed in Table 1	Theory, evidence, and gap identification

Table 3. Methodology hierarchy.

The architectural prototype (Section 8) provides a fourth, demonstrative layer: it does not claim to validate the hybrid pathway against real ABDM infrastructure, but shows, using synthetic data, what a resilient fallback could look like and what its effect on measured continuity might be.

8. Prototype and Reproducibility

An open-source prototype accompanies this proposal at [github.com/\[your-username\]/abdm-carebridge](https://github.com/[your-username]/abdm-carebridge). It generates synthetic, FHIR-shaped patient records, simulates the nine-stage journey in Table 2 under both an ABHA-only baseline and the proposed hybrid pathway, and visualizes the result as a continuity heatmap. Across an initial synthetic run of 10 patients, the baseline (ABHA available) scenario averaged 3.0 continuity breaks per journey, against 7.2 for the no-ABHA scenario – a preliminary, illustrative signal consistent with the proposal's central argument, not a validated empirical finding. All code, data generation scripts, and the dashboard are openly available and documented, including an explicit limitations statement.

9. Expected Outcomes

- A policy brief identifying specific structural gaps between ABDM's design assumptions and the operating conditions of Kerala's interstate informal migrant workforce.
- A documented, open-source architectural prototype demonstrating a privacy-preserving fallback pathway for continuity of care.
- Concrete, feasibility-scoped recommendations for Kerala's health and migrant welfare departments regarding ABHA onboarding, consent mechanisms, and cross-state data portability.

10. Relevance to Kerala's Planning Process

Kerala already demonstrated, during its COVID-19 response, the institutional capacity and intersectoral coordination required to deploy digital public health infrastructure for highly mobile populations, including migrant workers (Ummer et al., 2021). Extending that same capacity toward routine, ABDM-aligned continuity of care is an incremental institutional step rather than a new undertaking. This directly supports the state's migrant welfare policy and its ongoing e-Health Kerala – ABDM integration strategy.

11. Timeline

Weeks	Focus
1–2	Policy document review; finalize conceptual framework and data schema
3–4	Build synthetic FHIR generator and 9-stage journey simulator
5–6	Implement privacy-preserving linkage fallback; run comparative scenarios
7	Build dashboard visualization; compile administrative indicators
8	Write policy brief; finalize documentation, limitations statement, and repository

Table 4. Eight-week internship timeline.

12. Limitations

This study relies on synthetic data and secondary/administrative sources rather than primary fieldwork with migrant workers, given the two-month timeframe; it is scoped as a systems and policy analysis rather than an empirical adoption study. A full account of methodological and technical limitations is maintained separately in the project repository (*docs/limitations.md*) and updated as the work progresses.

References

- Bangla, S., Kapoor, A., & Jeer, G. (2026). Toward equitable digital health: An integrated framework addressing exclusion, ethics, and implementation across healthcare systems. *International Journal for Equity in Health*.
- Gandhi, A. P., & Soundappan, K. (2024). Perception towards electronic health records & uptake of digital health IDs among the urban residents in northern India: A mixed methods study. *Indian Journal of Medical Research*.
- Jayathissa, P., & Hewapathirana, R. (2023). Enhancing interoperability among health information systems in low- and middle-income countries: A review of challenges and strategies. *arXiv:2309.12326*.
- Mantri, N., et al. (2024). National digital health platform design using the WHO-ITU Digital Health Platform Handbook. *Frontiers in Digital Health*.
- National Health Authority. National Digital Health Blueprint. Ministry of Health & Family Welfare, Government of India (2019).
- Ngo, C., et al. (2024). RECONNECT: A federated, privacy-preserving record linkage architecture for fragmented health systems. *arXiv:2410.13880*.
- Raj, G. M., Dananjayan, S., & Agarwal, N. (2023). Inception of the Indian Digital Health Mission: Connecting the dots. *Health Care Science*.
- Sreekumar, S. (2023). Understanding Dalit equity: A critical analysis of primary health care policy discourse of Kerala in the context of Aardram Mission. *International Journal for Equity in Health*.
- Ummer, O., Scott, K., Mohan, D., et al. (2021). Connecting the dots: Kerala's use of digital technology during the COVID-19 response. *BMJ Global Health*, 6(Suppl 5).